Dan Munyaneza

Data Scientist | Analytics Engineer | Insights Strategist
AI & Machine Learning | Dashboarding & KPI Systems | Statistical Modeling
Data Visualization & Storytelling | Business Intelligence (BI)

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PROFESSIONAL SUMMARY

I am a **Data Scientist and Consultant** focused on turning data into practical solutions for better decision-making. I design **analytics products, KPI systems, interactive dashboards, statistical models,** and **end-to-end data workflows** using **Python, R, SQL, and Power BI,** including pipelines, **data warehouses,** and leadership-ready reporting.

My work with the **City of Kigali, Rwanda ICT Chamber,** and **NISR** transforms operational and survey data into meaningful insights for **youth programmes, urban digital services, gender-data initiatives,** and **evidence-based policy.** I combine technical delivery with a clear view of users and business needs, with a focus on **data quality, usability,** and **measurable impact.**

KEY ACHIEVEMENTS

- **Data Scientist Consultant** for **City of Kigali** digital programmes, including YEGO MIS, KSESM Smart Parking, and related analytics products.
- Advancing **GDDP** (Gender Data Lab) with **NISR** to make national gender survey data discoverable and actionable for advocacy and policy.
- 1st Place, **Umurava AI Hackathon 2025** (Umuranga AI): NLP/ML hiring intelligence reducing screening time by **70%+.**

EDUCATION

M.Sc. in Data Science, University of Rwanda (ACE-DS) | Starting 2026

African Center of Excellence in Data Science (ACE-DS) | Kigali, Rwanda
Focus: advanced analytics, statistical learning & applied data science research

B.Sc. (Honours) in Applied Statistics (Economic Statistics) | 2021–2024

University of Rwanda, UR | Kigali
Coursework: Econometrics, Statistical Modelling, Survey Methods, Data Analysis

SKILLS

Data Science, Engineering & Processing:

Python, R, SQL · Apache Spark, Hadoop, Kafka, Flink, Databricks · ETL frameworks: dbt, Airflow, Glue, Talend, SSIS

Cloud Platforms & DevOps:

AWS, Azure, Google Cloud: data storage, processing, and structured workflows · Git, Docker fundamentals

Database & Storage:

Relational: PostgreSQL, MySQL, SQL Server · NoSQL: MongoDB, Cassandra, Redis · Columnar: Redshift, BigQuery, Parquet, Delta Lake

Data Analytics & Visualization:

R Shiny, Tableau, Power BI, Plotly, advanced Excel · KPI tracking, predictive modeling, time series analysis, leadership-ready reporting

Languages:

Kinyarwanda (Native) · English (Fluent) · French (Intermediate)

PROFESSIONAL EXPERIENCE

Data Scientist Consultant, City of Kigali

Kigali, Rwanda | Present

- Provide data science consulting for **City of Kigali** digital programmes, turning operational and service data into leadership-ready KPIs and decision support.
- Design and deliver analytics products including **YEGO MIS** and **KSESM Smart Parking,** covering pipelines, validation, dashboards, and audit-ready reporting.
- Partner with programme, product, and engineering teams to define indicators, strengthen data quality, and improve service-delivery visibility across city platforms.
- Advise on governance-aligned data handling so city datasets remain reliable for monitoring, coordination, and continuous improvement.

Data Scientist, Rwanda ICT Chamber

Kigali, Rwanda | Present

- Support ICT Chamber digital and analytics initiatives with applied data science, reporting frameworks, and stakeholder-ready insight products.
- Build and refine analytical workflows using **Python, R, SQL,** and dashboard tools to inform programme performance and partner coordination.
- Collaborate across technical and non-technical teams to translate business and policy questions into measurable metrics and clear visualisations.

Data Scientist Consultant, NISR (Gender Data Lab / GDDP)

Kigali, Rwanda | Present

- Consult on the Rwanda Gender Data Discovery Platform (**GDDP),** enhancing how national gender-related survey data is discovered, interpreted, and used for evidence-based advocacy.
- Strengthen catalogue usability, sector dashboards, and data-quality signalling so CSOs, policymakers, and researchers can move from fragmented PDFs to actionable gender insights quickly.
- Contribute to platform enhancements that improve search, filtering, preview, and multi-format export, reducing time-to-insight for gender-focused programmes and policy work.
- Align analytics delivery with **NISR** survey assets (including DHS, EICV, Census, LFS and related series) and partner needs under the FAST programme ecosystem.

Data Science & Analytics Contributor, Rwanda Decoded

Kigali, Rwanda | Present

- Conduct multi-sector analysis across agriculture, education, and economics using **Python, R, SQL,** and Excel, producing reproducible notebooks and stakeholder-ready insight packs.
- Design dashboards and narrative analytics that help programme teams move from raw tables to clear recommendations for targeting, monitoring, and learning.

Data Analyst, Fortebet Rwanda

Kigali, Rwanda | Feb 2024 – Oct 2025

- Delivered transaction analytics covering trend detection, anomaly monitoring, cohort behaviour, and KPI reporting for operations and finance stakeholders.
- Engineered reproducible cleaning and validation workflows that reduced operational risk from incomplete, duplicated, or inconsistent transactional data.

Data Analytics Intern, Kiziguro District Hospital

Rwanda | Apr 2023 – Aug 2023

- Developed Excel and **Power BI** dashboards for hospital KPI monitoring, service throughput, and operational reporting used by facility managers.

PROJECTS

Project 1: YEGO & Incubation Centers MIS, City of Kigali

"Lead Analytics & Platform Delivery · City of Kigali"



Figure: "YEGO / Incubation Centers MIS: operational KPI dashboard"

The **YEGO & Incubation Centers Management Information System** is a centralized digital platform built to give **City of Kigali** programme managers real-time visibility into youth centre operations, incubation activity, and partner coordination. My data science role was to design the **analytics stack:** ingest operational records, clean and model them in a governed **PostgreSQL** store, and expose a **KPI layer** so visitor flows, attendance, programme participation, and service uptake could be monitored from one source of truth instead of fragmented spreadsheets.

I built **ETL workflows** in **Python and SQL** that extract centre transactions, validate completeness and duplicates, and load **fact tables** for attendance, visitors, and programme activity, with dimensions for centre, gender, date, and service type. On top of that **warehouse** I defined indicator logic (daily attendance trends, gender distribution, participation rates, and centre-level performance) and delivered interactive **leadership dashboards** so managers can spot bottlenecks, target underperforming centres, and produce **audit-ready reports** for coordination meetings and partner reviews.

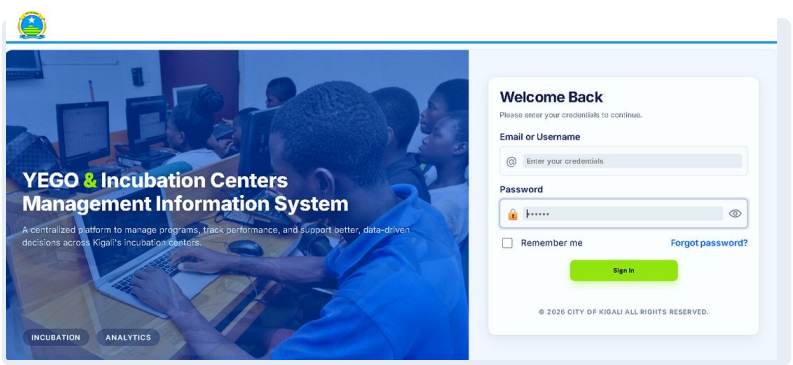
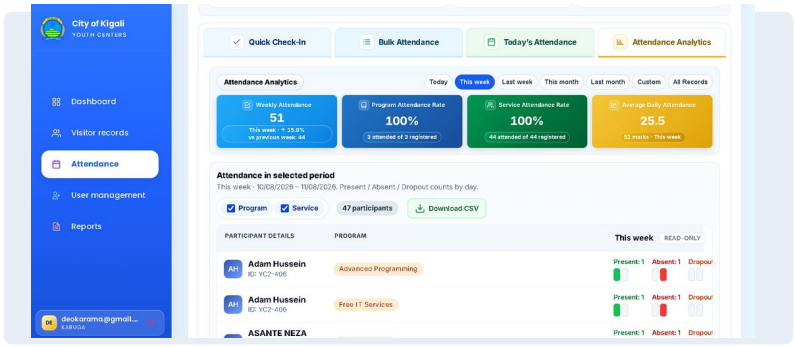


Figure: "Youth Center Management System: landing & secure sign-in"

Beyond reporting, the product experience supports secure staff access and day-to-day centre operations. I used **Python** and **SQL/PostgreSQL** for data preparation and quality control, then **Power BI** and **web dashboards** for decision consumption, so operators capture clean events at source and leadership reads the same governed metrics. Delivery followed version-controlled iteration aligned to **City of Kigali** digital service standards.

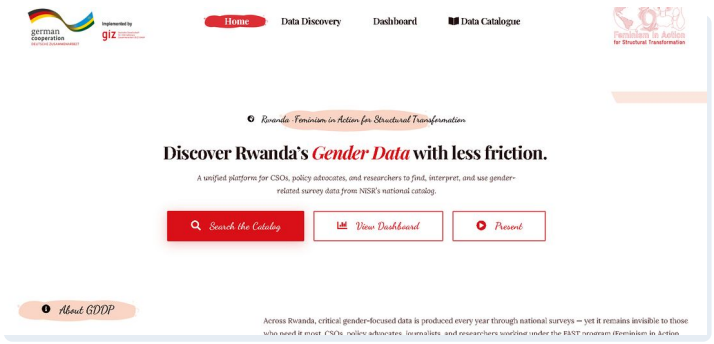


The attendance module is an **analytics product** on that **pipeline:** check-in events are stored, aggregated, and visualised so City teams can compare centres, track engagement over time, and improve **programme delivery** with evidence rather than anecdote. I focused on reliable event capture, reusable **SQL metrics,** and **dashboard** views that stay decision-ready for operations and reporting.

Figure: "Attendance check & analysis module supporting centre operations"

Project 2: GDDP – Rwanda Gender Data Discovery Platform (GIZ, NISR, MINICT)

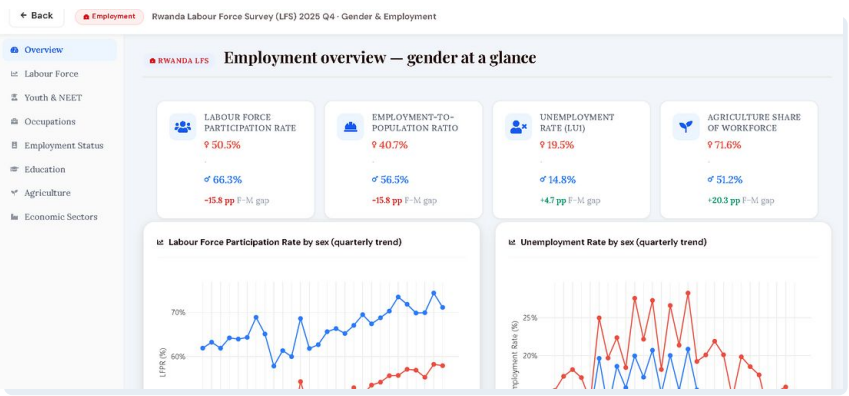
"Data Scientist Consultant · NISR · Gender Data Lab enhancement"



As **Data Scientist Consultant** with **NISR** on the **Gender Data Lab,** I contribute to enhancing the **Rwanda Gender Data Discovery Platform (GDDP).** The work is a data product: take national gender-related surveys, structure metadata and quality signals, and make the collection discoverable for civil society, policymakers, researchers, and journalists. Under the **FAST** program, GDDP acts as a **governed catalogue** and **analytics layer** on **NISR** survey assets rather than a pile of disconnected PDFs.

Figure: "GDDP Welcome page"

Before GDDP, gender data sat in scattered repositories, often PDF-heavy and hard to query. I help shape the **discovery pipeline:** connect users to **73 national surveys** produced by **NISR** (1978–2024), attach **quality scores** and gender relevance ratings, and organise the catalogue so CSOs, advocates, and researchers can judge dataset usability before they download. That scoring and metadata layer is what turns a file archive into a working **data warehouse** for advocacy.

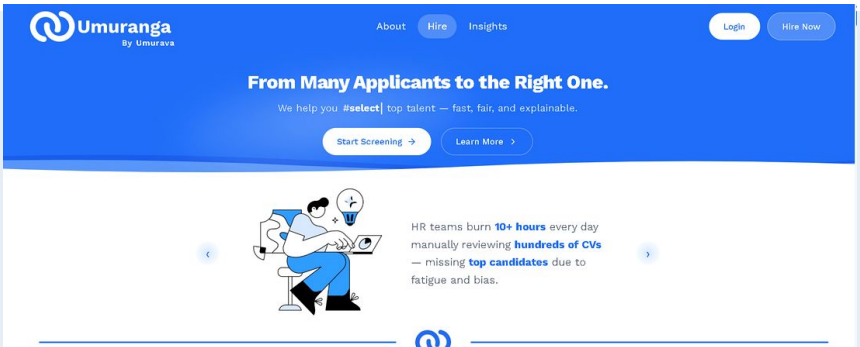


Sector dashboards sit on that catalogue: **gender-disaggregated** views from DHS, EICV, Census, LFS and related series, with live preview, filtering, and multi-format export. My consulting focus is the analytics value, search, and quality signalling so users can trust **NISR** gender data and move from question to insight in **under a minute.** The product work is **pipeline plus dashboard:** clean metadata in, decision-ready gender indicators out.

Figure: "GDDP"

Project 3: Umuranga AI, HR Intelligence Platform

"1st Place · Umurava AI Hackathon 2025"



Umuranga AI is an AI-assisted hiring intelligence platform that uses **NLP and machine learning** to screen, rank, and explain candidate fit with transparent scoring for recruiters. As lead builder for the winning **Umurava AI Hackathon** solution, I designed the full **data science workflow:** document ingestion, text cleaning, **feature extraction** from CVs and job descriptions, model-assisted ranking, and a recruiter interface that makes model outputs actionable rather than black-box scores.

I implemented **Python pipelines** for batch scoring, similarity and ranking features, and evaluation so the system could reduce early-stage screening effort by **70%+** while keeping **fairness** and **explainability.** Recruiter dashboards surface why candidates were ranked, which lets HR teams focus human review on high-signal profiles. Delivery followed production-minded practices: reproducible preprocessing, versioned models, and metrics suited to high-volume hiring operations.

Project 4: Genda, Smart Urban Mobility Platform

"Collaborator · City of Kigali · Ongoing (NST2 Pillar 3.6)"

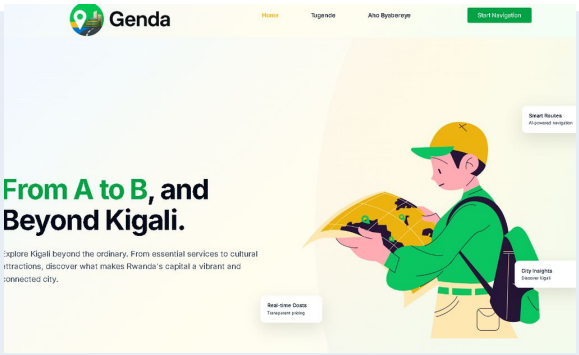


Figure: "Genda: smart navigation and multimodal mobility experience"

Genda is a smart urban mobility initiative developed in collaboration with **City of Kigali** to improve navigation, trip planning, and multimodal transport insights across the city. My **data science** work covers **movement data:** trip and routing records, transport-cost signals, and decision-ready views of how people move by moto-taxi, bus, and greener options. I contribute **pipelines and analytics** that turn raw mobility events into maps, indicators, and product insights for city programmes.

Under **NST2 Pillar 3.6** the analysis links real-world mobility behaviour to **service-delivery** and policy questions: where friction occurs, which corridors are overloaded, and how digital navigation can support safer, more efficient, and more inclusive urban transport. I focus on scalable **data models,** reusable metrics, and **dashboards** that stay actionable for both citizens and city programme teams.

Project 5: Smart Parking & Service Delivery System (KSESM)

"City of Kigali, Digital Public Services Platform"

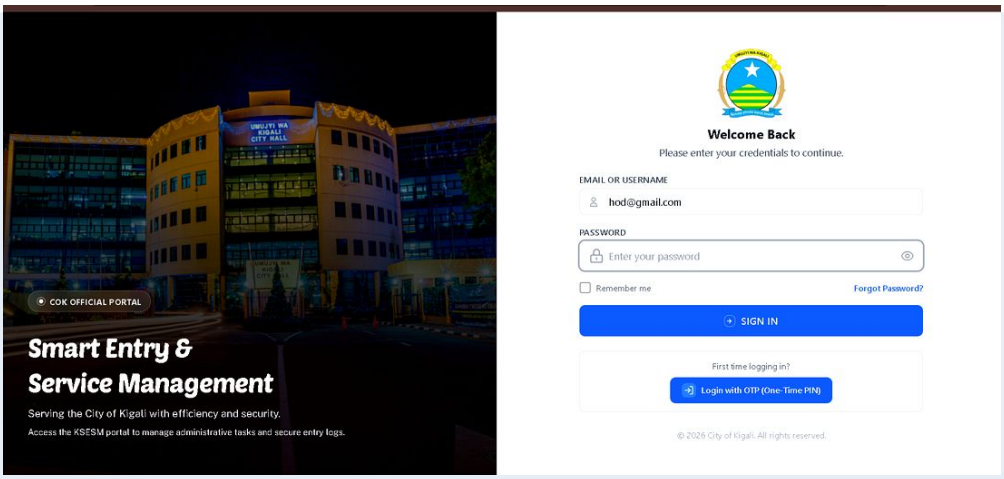


Figure: "KSESM Smart Parking: citizen-facing service portal"

The **KSESM Smart Parking & Service Delivery System** modernizes **City of Kigali** parking operations and related public-service workflows through a secure digital portal. My data science contribution is the operational data layer: capture gate and parking events, run **validation pipelines,** store them in a structured **warehouse,** and serve **KPI dashboards** for entry management, administrative processes, and citizen-facing service access, so teams move from

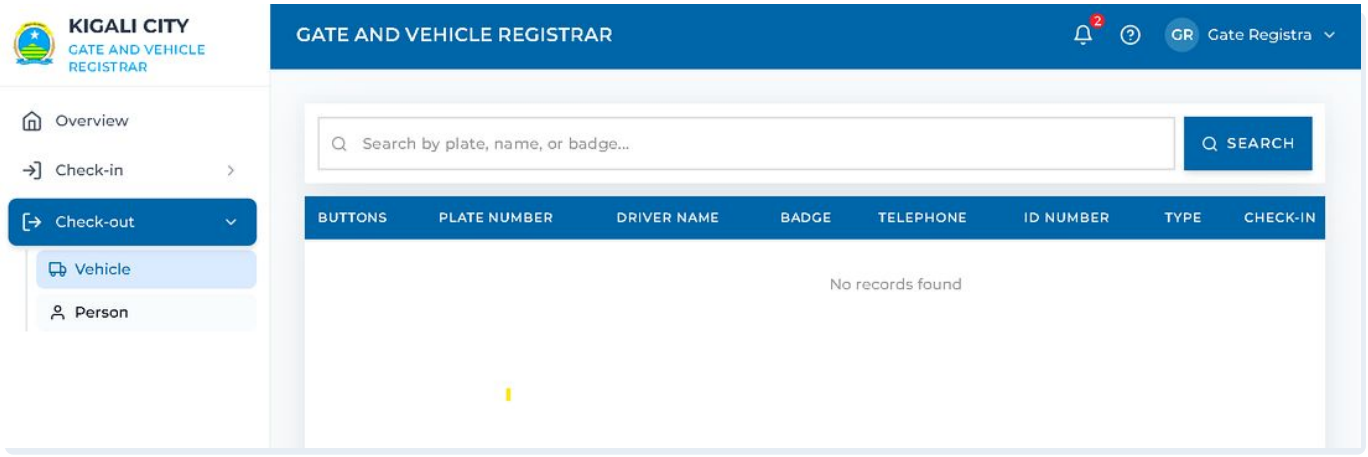


Figure: "Secure vehicle & person checkout / gate registry workflow"

The gate registry module is the capture point for that **pipeline:** officers check vehicles and persons in and out, search by plate, name, or badge, and write structured events into the operational store. I designed the **data model** and reporting logic for **throughput,** dwell, **compliance,** and service-quality monitoring, then exposed those metrics in **dashboards** so city teams can supervise gates and parking activity with the same governed records used at the counter.

CERTIFICATIONS

- **Google Cloud Professional Data Engineer**
  Google Cloud
- **Data Analysis & Visualization**
  Rwanda ICT Chamber
- **Databricks Certified Associate**
  Databricks
- **Data Science (Advanced)**
  ICT Chamber

RESEARCH & PUBLICATIONS

Impact of Economic Factors on Land Price in Rwanda

BSc dissertation analyzing how key economic variables influence land prices in Rwanda, strengthening skills in **econometric analysis, data visualization,** research writing, and **policy evaluation.** The study applied statistical modelling and visual storytelling to connect experimental evidence with practical implications for land markets and development planning.

REFERENCES

- Mr. Gabriel Nduwayezu, Chief Digital Officer (CDO), City of Kigali · Tel: +250 788 701 492 · gnduwayezu@kigalicity.gov.rw
- Mr. Nsabimana Emmanuel, Project Manager, Rwanda ICT Chamber · Tel: +250 786 958 109 · emmanuel@ictchamber.rw
- Dr. Idrissa Kayijuka, Head of Department, Applied Statistics, University of Rwanda · Tel: +250 788 665 069 · kayijukai@gmail.com

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